

EL ARISH INTL, Egypt

WMO: 623370

Lat: 31.073N

Lon: 33.836E

Elev: 32

StdP: 100.94

Time Zone: 2.00 (E02)

Period: 95-17

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	6.8	-2.2	3.2	14.7	-0.1	3.8	15.0	10.3	14.0	9.2	14.0	2.0	160	0.334

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.8	35.1	21.7	33.5	23.0	32.3	23.5	26.3	30.9	25.7	30.5	25.1	30.1	3.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.1	20.3	28.9	24.2	19.2	28.5	23.6	18.5	28.2	82.3	31.0	79.4	30.6	76.9	29.9	33.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.5	5.5	DB	2.5	41.3	1.6	2.3	1.4	43.0	0.5	44.3	-0.4	45.6	-1.5	47.3	(4)
(5)			WB	1.5	28.2	2.0	1.8	0.0	29.5	-1.1	30.5	-2.2	31.6	-3.6	32.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	20.6	13.4	14.3	16.2	19.0	21.8	24.7	26.9	27.3	25.8	23.1	19.1	15.1	(6)	
	DBStd	5.32	2.15	2.41	3.02	3.13	2.52	1.87	1.21	1.14	1.52	1.96	2.39	1.96	(7)	
	HDD10.0	2	2	1	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	498	153	117	83	24	1	0	0	0	0	0	18	101	(9)	
	CDD10.0	3868	108	120	192	269	367	441	523	537	474	404	274	159	(10)	
	CDD18.3	1322	1	3	17	44	109	191	265	279	224	146	41	1	(11)	
	CDH23.3	13263	16	43	201	440	959	1857	2919	3153	2254	1133	263	25	(12)	
(13)	CDH26.7	5186	2	11	88	196	316	681	1282	1402	856	297	52	3	(13)	
(14)	Wind	WSAvg	2.2	2.6	2.8	2.6	2.5	2.2	2.0	1.9	1.7	1.9	1.8	2.0	2.2	(14)
(15) Precipitation	PrecAvg	117	30	19	18	4	1	0	0	0	0	6	13	23	(15)	
	PrecMax	209	69	52	61	16	6	0	0	0	1	41	76	69	(16)	
	PrecMin	45	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	45	19	12	14	4	1	0	0	0	0	9	17	20	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.2	28.2	34.9	36.8	36.4	36.7	36.0	35.4	36.2	34.2	31.0	26.4	(19)	
		MCWB	14.4	15.7	17.1	18.1	18.7	21.2	23.2	25.2	22.0	20.9	17.4	14.7	(20)	
	2%	DB	22.0	24.0	28.8	32.2	32.3	33.1	33.9	33.7	32.9	30.8	27.9	23.2	(21)	
		MCWB	13.0	14.2	16.0	17.1	19.2	21.7	24.2	25.1	23.7	22.5	19.1	14.9	(22)	
	5%	DB	19.9	21.1	24.5	28.1	29.8	31.3	32.8	32.6	31.5	29.2	25.9	21.6	(23)	
		MCWB	13.6	13.5	15.6	16.6	19.2	22.5	24.3	24.9	23.9	22.1	18.5	14.6	(24)	
	10%	DB	18.4	19.3	21.8	24.9	27.9	30.1	31.8	31.9	30.5	28.2	24.5	20.3	(25)	
		MCWB	12.9	13.2	14.9	16.7	19.4	22.4	24.2	24.6	23.3	21.6	18.2	14.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.8	17.2	19.1	19.7	22.6	25.0	26.7	28.1	27.0	25.2	22.4	18.1	(27)	
		MCDB	20.4	23.5	26.1	27.8	28.5	30.1	31.3	32.1	30.7	29.1	25.9	21.8	(28)	
	2%	WB	15.3	15.7	17.6	18.7	21.6	24.2	25.9	26.6	25.5	24.1	21.0	16.7	(29)	
		MCDB	19.2	20.7	24.5	26.7	27.0	29.5	31.0	31.1	30.2	28.5	24.8	20.4	(30)	
	5%	WB	14.4	14.7	16.7	18.0	20.9	23.6	25.3	26.0	24.7	23.2	19.9	15.8	(31)	
		MCDB	18.5	19.2	22.0	24.8	26.4	28.9	30.5	30.7	29.7	27.6	23.9	19.7	(32)	
	10%	WB	13.4	13.9	15.8	17.3	20.3	23.0	24.8	25.4	24.0	22.3	18.9	15.0	(33)	
		MCDB	17.6	18.2	20.6	23.5	25.8	28.1	30.0	30.2	29.0	26.9	23.1	19.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	10.7	11.3	11.9	12.0	11.1	10.2	9.8	10.1	10.7	11.4	11.3	(35)	
		MCDBR	13.3	14.4	16.5	17.9	15.9	13.6	12.0	10.7	11.5	12.2	13.1	13.3	(36)	
	5% WB	MCWBR	7.3	6.8	7.0	6.7	5.9	5.1	4.8	4.1	4.6	5.7	6.6	7.4	(37)	
		MCWBR	11.9	12.3	13.3	14.1	12.6	11.6	10.8	10.1	10.5	11.2	11.2	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.380	0.416	0.460	0.504	0.501	0.426	0.432	0.434	0.436	0.446	0.394	0.378	(40)	
		taud	2.169	2.052	1.933	1.840	1.879	2.125	2.124	2.125	2.114	2.070	2.192	2.195	(41)	
	Ebn at Noon		826	829	820	799	803	862	855	850	832	788	804	805	(42)	
		Edh at Noon	123	151	183	208	202	157	157	155	150	147	119	114	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.23	4.09	5.41	6.49	7.35	7.98	7.68	7.06	6.05	4.70	3.59	2.99	(44)	
	RadStd		0.14	0.21	0.24	0.23	0.26	0.12	0.12	0.12	0.15	0.18	0.13	0.13	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (15 neighbors)	+0.46	N/A	-2.08	N/A	N/A	N/A	N/A	-60	+142	+105	(47)

Nomenclature: See separate page